Explained or Certified? Examining the Influence of XAI and AI-Seals on Users' Trust and Understanding

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Abstract

Explainable artificial intelligence (XAI) aims to increase users' understanding and trust in AI-powered systems. However, while XAI has shown benefits, it also comes with shortcomings such as shifting the cognitive burden to users. Certifications, so-called AI-Seals, pose an alternative (or addition) to XAI by signaling a system's trustworthiness through a label. Despite their potential, empirical investigations of AI-Seals remain scarce. In an online experiment (N = 436), a 2 (XAI vs. no XAI) x 2 (AI-Seal vs. no AI-Seal) between-subject design was conducted to analyze the effects of XAI, AI-Seals, or a combination of both on users' perceived and factual understanding and trust in the AI-application. Results show that XAI led to more perceived and factual understanding but had no effect on trust. Against our assumptions, an AI-Seal could neither promote perceived understanding nor trust. Likewise, the combination did not increase trust but has potential to increase factual understanding.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Empirical studies in HCI**;

Keywords

Artificial Intelligence, Explainable-AI, AI-Seal, Trust, Understanding

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1 Introduction

In recent years, artificial intelligence (AI) has gained increasing relevance and is being applied across a wide range of domains. Due to rapid technological advances, the performance of AI systems has reached a level where even experts may struggle to fully comprehend their internal processes [15]. However, since the majority of AI-users do not have detailed knowledge about AI, it is not surprising that their perception of AI systems is primarily that of a black-box that cannot be trusted, leading to the "black-box problem" (the inability to understand the inner workings of an AI due to a lack of transparency) which can foster mistrust and reduced acceptance of AI in society [40]. To counteract this challenge, effective strategies are needed that promote transparency, enhance understanding, and strengthen user trust in AI.

One method to counter AI's intransparency is explainable artificial intelligence (XAI), which aims to make the decision-making processes of AI more transparent by providing human-understandable explanations. These explanations are intended to help users form accurate mental models of how an AI system functions and thereby strengthen trust [9, 10]. Empirical studies have shown that XAI can improve both perceived and factual understanding, and, in many cases, also lead to increased trust in AI systems [9, 39]. In addition, by promoting understanding, the resulting trust is more likely to be appropriate – meaning that it matches the actual trustworthiness and reliability of the AI system [44]. However, other research points to potential downsides: if explanations are too complex or poorly matched to users' cognitive resources, they may hinder rather than promote understanding and trust [32].

An alternative (or addition) to promote users' trust is the certification of AI's trustworthiness by independent institutions [41]. Through AI-Seals of trust, these certifications are intended to communicate to users that the system complies with legislative and industry requirements. First empirical findings suggest that AI-Seals can promote trust, particularly when the certifying institution is perceived as credible, and the context involves high stakes [36, 43]. However, concerns remain that such seals may foster a superficial sense of assurance or even blind trust if users rely on them heuristically without deeper engagement [18].

This raises the question of whether a combination of XAI and AI-Seal could leverage the strengths of both approaches while compensating for their individual weaknesses. The goal of this study is, therefore, to investigate to what extent XAI and an AI-Seal – each separately and in combination – can contribute to increasing users' understanding of and trust in AI systems.

2 Related Work

2.1 The Role of Factual and Perceived Understanding in Context of System-like Trust

To establish trust in AI, it is essential to comprehend how understanding contributes to trust. Lee and See's [23] concept of trust highlights the significance of understanding, which reduces uncertainty and vulnerability, enabling a better judgment about the trustworthiness of a partner. With regard to AI systems, this means that better understanding would lead to a better assessment of the reliability of AI and to more trust [10].

In order to examine this relationship more precisely, it is crucial to recognize that the concept of understanding is multifaceted and can be differentiated into two distinct forms, factual understanding and perceived understanding. Perceived understanding describes subjectiveknowledge and is the feeling of having understood something. Factual understanding, describes the factual (objective) recall about a topic [24]. While previous findings led to the suggestion that perceived and factual understanding are two different constructs [6], other studies show that they can correlate with each other [39]. Therefore, the question arises what kind of understanding could be increased by the trust-building measures used in this study and whether this reduces uncertainty and vulnerability regarding AI and therefore enhances trust in it [23]. To address this, the present study examines both perceived and factual understanding, as they capture complementary aspects of users' comprehension and may differentially contribute to the formation of trust.

Yet, in order to fully understand how trust in AI can be effectively promoted, it is not only important to identify ways of increasing it, but also to examine the factors that influence its development, as "trust has become a central variable to explain both resistance to use automated systems (disuse) as well as overreliance on automated systems (misuse)" [44, p. 2]. Various measurements have been proposed to evaluate trust in AI systems, e.g. system-like trust, which consists of the features reliability, functionality and helpfulness [21]. Reliability describes the technology operates properly at any time, functionality means technology has the required functions to fulfill the tasks for which it was developed and helpfulness refers to the belief that proper and responsive help is offered by technology to the user [21, 27]. According to Lee and See [23] performance, process and purpose are "the general bases of trust" (p. 59) and can be adapted to the features of system-like trust. Thereby, the factor process represents reliability, performance can be equated with functionality, and the equivalent to the factor purpose is helpfulness. Therefore, in this study system-like trust was chosen to measure the construct trust.

Taken together, understanding and trust form the conceptual foundation of this research. In order to investigate how these two constructs can be actively influenced, two different approaches in the context of human-AI interaction – XAI and AI-Seals – are presented below and analyzed with regard to their potential to promote user understanding and strengthen trust in AI systems.

2.2 From Explainable Artificial Intelligence to AI-Seals of Trust

XAI is one of the most widely discussed approaches to address the black-box character of AI systems and to foster user trust and understanding. It encompasses a set of techniques and various methods that aim to make the decision-making processes of AI systems more transparent and interpretable for human users by providing insights into AI decision-making processes [11, 13]. These methods can be categorized based on their approach, applicability, and the type of explanation they offer. Key methods include model agnostic (e.g., LIME [35]) vs. model specific methods, global vs. local explanations, and factual (clarifying the reasons behind specific outputs), vs. contrastive explanations (clarify why alternative outputs were not produced) [17, 20].

Empirical findings confirm the promise of XAI. Prior studies have demonstrated that explanations in general can enhance user comprehension, increase perceived transparency, and contribute to higher levels of trust in AI systems [9, 16, 37]. Among the different types of explanations, this study focuses on contrastive explanations, as recent research has identified that contrastive explanations were particularly effective in fostering both factual and perceived understanding, due to their alignment with human explanatory patterns and their demonstrated impact on improving user mental models and trust [30, 39, 42].

However, XAI is not without limitations. Recent critiques highlight that explanations can become overly technical or cognitively demanding, especially for non-expert users, thereby reducing their accessibility and effectiveness [8, 32]. In addition, Bertrand et al. [2] found that users' cognitive biases can shape how XAI is interpreted, potentially reducing its utility in decision-making contexts. Similarly, Miller [28] argues that explanation in AI may shift the cognitive burden to the user, who is then responsible for making sense of complex output.

Given these limitations, it is necessary to consider additional or complementary strategies to support (appropriate and calibrated) trust in AI – particularly for diverse user groups and real-world use cases. One such approach could be a certification of AI systems in the form of an AI-Seal [36, 41, 43], since seals have long been used in other domains to foster trust. In line with this, the results of Jiang et al. [14], show that the presentation of third-party seals in an online shopping context positively influences trust in e-marketers. Contrary, Pennington et al. [33] show that the presentation of a seal does not lead to significantly more trust in Business-to-Consumer transactions. However, despite the heterogeneous results, theoretical considerations assume that AI-Seals still could increase trust towards AI (e.g., [19, 31]).

Recent empirical studies specifically investigating AI-Seals further refine this assumption. Wischnewski et al. [43] found that while AI-Seals generally did not affect participants' trust directly, participants' trust in the AI system increased if they trusted the seal-issuing institution. Moreover, although participants understood verification seals the least, they expressed the highest desire for AI system verification. Similarly, Scharowski et al. [36] demonstrated that certification labels significantly increased end-users' trust and willingness to use AI in both low- and high-stakes scenarios, with stronger effects observed in high-stakes contexts.

In contrast to their potential for increasing trust, the role of AI-Seals in enhancing user understanding remains uncertain. López Jiménez et al. [25] suggest that seals can enhance understanding especially in areas that are difficult for consumers to access, which could also apply to AI. However, other research argues that this effect is limited, as seals often provide little background information [5], making an increase in factual understanding improbable. Instead, it is more likely that seals merely create an "impression of understanding" – in other words, increase perceived understanding – with users assuming they have received meaningful information when this is not the case. This illusion is likely driven by heuristic, rather than systematic, information processing [44]. Therefore, further research is needed to clarify whether AI-Seals can effectively contribute to genuine user understanding, rather than merely creating a superficial impression of comprehension.

Given that XAI and AI-Seals are grounded in fundamentally different mechanisms of user engagement – one rather aiming at cognitive elaboration, the other relying on heuristic cues – a combined application appears theoretically promising.

2.3 Combining Explainable Artificial Intelligence and AI-Seals

While both XAI and AI-Seals offer promising strategies to increase trust and understanding in AI systems, each comes with limitations that could be addressed through a combined approach. The idea behind this combination is to positively complement both methods to maximize the advantages (increased trust, understanding, transparency [9, 39]) of each, while minimizing the disadvantages (e.g., blind trust, false understanding [8, 18]).

To theoretically frame this idea, the Elaboration Likelihood Model (ELM) [34] offers a useful lens. The purpose of the ELM is "organizing, categorizing, and understanding the basic process underlying the effectiveness of persuasive communication" [34, p. 3] and the model distinguishes between two types of information processing: the central route, which involves deep, effortful cognitive engagement, and the peripheral route, which relies on superficial cues and heuristics. Importantly, the selection of a processing route is not determined by the nature of the message itself, but by the receiver's motivation and ability to elaborate. High motivation and sufficient cognitive resources promote central-route processing, while low motivation or limited ability tend to activate peripheral-route processing [34].

Using this theoretical foundation, the present study considers whether the presence of explanatory content, such as XAI, might serve as a system affordance that encourages central-route processing. Due to the complexity and cognitive demand of many explanations, it appears plausible that XAI can stimulate more central-route processing and thus foster higher levels of understanding and trust. However, recent research also challenges this notion. Buçinca et al. [4] found that even the mere presence of an explanation can serve as a heuristic, associated with peripheral-route processing, leading users to trust AI outputs without engaging in deeper reasoning. This suggests that the processing route elicited by XAI is not fixed but may vary depending on boundary conditions such as explanation quality, user expertise, or task context – alongside users' motivation and ability. Consequently, it remains an open and theoretically relevant question which processing route XAI actually triggers in practice.

In contrast, AI-Seals are more clearly aligned with peripheral-route processing [22] and serve as a peripheral cue, to assess the trustworthiness of a product or system [3]. Since they typically lack substantive information about the system's internal functioning, it can be expected that an AI-Seal may only lead to the feeling of understanding the AI-application, whereas no factual understanding will be gained.

Hence, while both interventions, XAI and AI-seals, may follow distinct processing routes, their combination may offer a complementary strategy that leverages the benefits of both. A joint presentation of XAI and AI-Seals could engage both the central and peripheral routes simultaneously – responding to diverse user preferences, cognitive capacities, and situational demands. Although Petty and Cacioppo's [34] ELM theory assumes that the route of processing is primarily determined by users' motivation and ability to elaborate, this raises the question of whether specific system features – such as the combination of explanations and heuristic cues – might nonetheless afford certain processing routes more strongly than others. In this case, a dual-route approach may foster greater trust and understanding by uniting the transparency and cognitive engagement of XAI with the credibility and intuitive reassurance of certification through an AI-Seal. However, this combined approach is not without potential downsides. Presenting both an explanation and a certification label could increase cognitive load, which in turn might reduce trust if users feel overwhelmed or confused [7]. Additionally, there is a risk that users selectively attend to only one of the cues – possibly defaulting to the more accessible heuristic path, especially under time pressure or low motivation.

Since such a combined application has received little empirical attention so far, this study seeks to explore whether the integration of XAI and AI-Seals can indeed maximize benefits while minimizing drawbacks in promoting understanding and trust in AI systems.

3 Method

3.1 Hypotheses and Research Question

Based on the theoretical considerations presented above, this study investigates how different trust-building measures – XAI, AI-Seal or the combination – affect users' understanding and trust in AI systems. In doing so, we distinguish between perceived understanding, factual understanding, and system-like trust as key dependent variables.

Combining both XAI and AI-Seals could potentially maximize their respective benefits. While XAI provides in-depth information about the system's logic, the seal may offer a simple, trust-enhancing cue. This leads to our primary research question:

RQ1: Which combination of trust-building measures in AI (XAI vs. AI-Seal vs. Combination of both) will elicit the highest levels of a) perceived, b) factual understanding and c) system-like trust?

Prior research has shown that XAI can improve users' understanding of AI systems by providing insight into the underlying decision-making processes. When well-designed, such explanations can enhance both perceived and factual understanding and may also contribute to system trust [39]. Accordingly, we propose the following hypothesis:

H1: Presenting XAI will elicit more a) perceived understanding, b) factual understanding and c) system-like trust compared to no presentation of XAI.

In contrast, AI-Seals are designed to serve as heuristic cues indicating that an AI system meets certain quality or safety standards. While they may not directly improve factual understanding, they may foster perceived understanding and increase trust by signaling credibility. Therefore, the second hypothesis is as follows:

H2: Presenting an AI-Seal will elicit more a) perceived understanding and b) system-like trust compared to no presentation of AI-Seal.

We additionally pre-registered hypotheses and a research question regarding the potential moderating roles of Need for Cognition and Propensity to Trust. As the corresponding results – reported in the supplementary material (OSF - Open Science Framework) – did not yield theoretically meaningful findings, they are not further addressed in the main manuscript.

3.2 Experimental Design and Procedure

The study was preregistered on the Open Science Framework (Preregistration Link) and received ethical approval from the ethics committee of the University of Duisburg-Essen. To explore the hypotheses and research questions, trust-building measures were manipulated and an online-experiment, using a 2 (XAI vs. no XAI) x 2 (AI-Seal vs. no AI-Seal) between-subject design was conducted. The group with no trust-building measure served as baseline. Participants were recruited through the online panel provider Prolific and randomly assigned to one of the four conditions. The data collection took place from January 27 to March 02, 2023, and included the interaction with an AI system and a questionnaire.

First, participants answered questions about their sociodemographic characteristics, Need for Cognition, and Propensity to Trust.

Afterwards, they were directed to the AI system, a recommendation system called the breakfast guide, which recommends breakfasts based on user preferences and restrictions. Beforehand, a cover story explained that the guide will be used for people with celiac disease, to highlight the perceived relevance of the guide. At the beginning of the interaction, the participants indicated how they would like to modify their breakfast by providing their preferences (e.g., taste, difficulty) and restrictions (e.g., allergies, no meat) and how important they rated each aspect (five-point Likert scale: 1 = *not important* to 5 = *important*).

The guide then provided recipe suggestions. In the XAI condition, participants received an explanation of the guide's recommendations (Figure 1), while participants in the AI-Seal condition were shown an AI-Seal (Figure 2). In the combination condition both XAI and AI-Seal were presented. Participants in the control group were only presented with the recipe suggestions. Afterwards, participants returned to the questionnaire and answered questions about factual understanding, perceived understanding, and system-like trust, followed by a manipulation check.

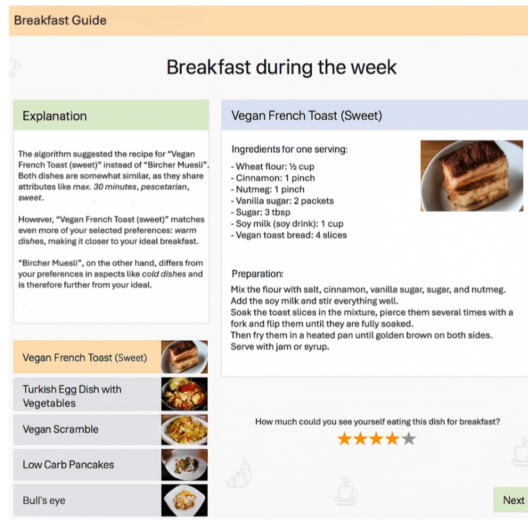

Figure 1: Recommendations for a breakfast with XAI

3.2.1 *Design of XAI and AI-Seal.* For the XAI a contrastive explanation was used (Figure 1). The self-designed AI-Seal (Figure 2) was based on existing conventions for the design of seals of approval, featuring a humanoid AI head and the word "certified" to indicate its authenticity. A fictional number, structured like a real code number according to DIN/ISO standard, reinforced the impression of certification. In addition to the presentation of the AI-Seal, participants were informed that the seal was certified by a consortium of reputable universities from the region, in order to enhance its perceived credibility. AI-Seal and XAI were placed in the same position for consistency.

Figure 2: Self-designed AI-Seal

3.2.2 Manipulation Check. A manipulation check was conducted to determine if participants had seen an AI-Seal. The initial data collection wave (N = 113) yielded a 43% failure rate on this check. To improve salience, the seal (Figure 2) was added to the check, additionally with the seal and a certification disclaimer in the briefing of the AI-Seal and combined condition. This revised procedure resulted in a pass rate of 85% for all participants sampled thereafter (N = 388). Data from all collection waves were combined for the final analysis.

3.3 Sample

G*Power was used to conduct a power analysis and determine a minimum sample size of 415 respondents (see preregistration). The study recruited 501 participants aged 18 or older with good understanding of German through the "Prolific" panel. 22 participants were excluded due to failing the attention check or having an extremely short or long completion time. 43 participants were excluded due to variance homogeneity violations and being identified as outliers based on a z-standardization of dependent variables. Z-values outside the range of -3.29 and 3.29 were removed [12]. The analysis involved 436 participants randomly assigned to four conditions: XAI (n = 111), A-Seal (n = 106), combination (n = 113), and control (n = 106). Of these participants, 220 (50.5%) were male, 214 (49.1%) female, and 2 (0.5%) identified as diverse. Their average age was 29.68 years (*SD* = 8.61), ranging from 18 to 63 years.

3.4 Measures

The items for the main scales analyzed in this paper are available in the supplementary material on the Open Science Framework (OSF - Open Science Framework).

3.4.1 Perceived and Factual Understanding. To examine perceived understanding about the breakfast guide, participants answered six items from Szczuka et al. [39]. These included two questions e.g. "Do you feel able to explain to other people how the algorithm works?" and four statements e.g. "I am confident that I understand the basic functions of the algorithm." which participants had to rate on a five-point Likert scale (1 = *Disagree* to 5 = *Agree*; M = 3.46, *SD* = 0.64; α = .79). Additionally, participants passed an attention check.

To measure factual understanding, participants rated eleven items from Szczuka et al. [39] as true or false (e.g. "The algorithm takes into account the importance of my specified preferences"). Participants accurately identified 8.24 statements on average (*SD* = 1.40), labeling wrong statements as "false" and true statements as "right.", while also indicating their level of response certainty for each statement on a 5-point Likert scale (1 = *Not certain at all* to 5 = *Very certain*; M = 3.65, *SD* = 0.58).

3.4.2 System-like Trust. System-like trust was measured by the items based on Szczuka et al. [39]. *Reliability* was measured using a modified version of the Perceived Reliability Scale of Madsen and Gregor´s [26], including three of the original five items, e.g. "The breakfast guide has responded to my specific input with specific recipes on different occasions respectively". The items were rated on a five-point Likert scale (1 = *Disagree* to 5 = *Agree, M* = 4.23, *SD* = 0.68; α = .71). *Functionality* was measured using three modified items from McKnight et al. [27] (e.g., "The breakfast guide has the functionality necessary for my tasks."; five-point Likert scale: 1 = *Disagree* to 5 = *Agree, M* = 3.89, *SD* = 0.89; α = .90). To measure *performance*, four adapted items from Shin's [38] Performance Scale were used (e.g., "The recipes recommended in the breakfast guide are generally accurate"; five-point Likert scale: 1 = *Disagree* to 5 = *Agree, M* = 3.70, *SD* = 0.82; α = .85).

4 Results

Calculations were performed using the statistical software SPSS 28. The significance level of α = .05 was used for all statistical tests. To gain a preliminary understanding of the data set, descriptive statistics of the dependent variables were calculated, summarized in Table 1.

Table 1: Descriptive Statistics of Dependent Variables by Condition and in Total

	XAI		AI-Seal		Combination		Control Group		Total	
	(n=111)		(n=106)		(n=113)		(n=106)		(N=436)	
	M	SD	M	SD	M	SD	M	SD	M	SD
Perceived understanding	3.59	0.61	3.34	0.61	3.55	0.64	3.35	0.66	3.46	0.64
Factual understanding	8.50	1.47	7.77	1.24	8.75	1.31	7.89	1.33	8.24	1.40
System-like trust	3.95	0.63	3.86	0.69	3.96	0.68	3.89	0.71	3.92	0.68

4.1 Hypotheses and Research Question Testing

Assumptions for each statistical procedure were checked before calculations. For MANOVAs (H1, H2, RQ1), multivariate normality was tested using Shapiro-Wilk tests (all $p < .001$) and Levene tests (all $p > .05$) were used to examine variance homogeneity. As MANOVA is considered robust to normal distribution violations, no countermeasures were taken.

For a clear overview of the main effects on each dependent variable, the results of the univariate ANOVAs are summarized in Table 2. The detailed post-hoc analyses examining the specific group differences are elaborated on in the following sections.

Table 2: Results of the Univariate Analyses of Variance (ANOVAs) on all Dependent Variables

	df	F	p	ηp^2
Perceived understanding	(3, 432)	4.62	.003	0.03
Factual understanding	(3, 432)	13.68	<.001	0.09
System-like trust	(3, 432)	0.57	.638	

4.2 Effect of XAI

To test H1, a MANOVA was conducted. It was hypothesized that presenting a XAI will cause higher levels of a) perceived, b) factual understanding, and c) system-like trust compared to no explanation. As dependent variables, the calculation considered system-like trust, perceived understanding and factual understanding, and the presence of XAI as a between-subject factor.

The MANOVA showed a significant difference between presenting a trust-building measure on the combined dependent variables, $V = 0.11$, $F(9, 1296) = 5.27$, $p < .001$. Post-hoc univariate ANOVAs were conducted for every dependent variable (see Table 2 for summary). Results showed a significant effect on perceived understanding, supporting hypothesis H1a, and a significant effect for factual understanding, supporting hypothesis H1b. There was no effect on system-like trust, which opposes H1c.

Tukey HSD post-hoc analysis on perceived understanding showed a significant difference between the control group and the XAI condition, $p = .028$ (*M_Diff* = 0.24, 95%-CI[0.02, 0.46]). A significant difference was also found for factual understanding, $p = .004$ (*M_Diff* = 0.62, 95%-CI[0.15, 1.09]), but not for system-like trust, $p = .920$ (*M_Diff* = 0.06, 95%-CI[-0.18, 0.30]). This indicates that people to whom the algorithm's procedure was explained via XAI had a significantly higher perceived understanding and factual understanding, but no significantly higher system-like trust compared to no presentation of XAI (Table 1).

4.3 Effect of AI-Seal

A second MANOVA was computed to test H2: the presentation of an AI-Seal elicits more a) perceived understanding and b) system-like trust than no presentation of an AI-Seal. System-like trust and perceived understanding were considered as dependent variables, and the presentation of an AI-Seal was included as a between-subject factor.

The MANOVA found a significant difference between presenting a trust-building measure on the combined dependent variables, $V = 0.03$, $F(6, 864) = 2.31$, $p = .032$. The results of post-hoc univariate ANOVAs for every dependent variable are summarized in Table 2. These results showed a significant effect on perceived understanding, supporting H2a. There was no significant effect on system-like trust, which contradicts H2b.

Tukey HSD post-hoc analysis on perceived understanding showed no significant difference between the control group and the AI-Seal condition, $p = 1.000$ (*M_Diff* = -0.01, 95%-CI[-0.23, 0.22]). Also in terms of system-like trust, no significant difference was found post-hoc between AI-Seal condition and control group, $p = 0.984$ (*M_Diff* = -0.03, 95%-CI[-0.28, 0.21]).

This implies that AI-Seals do not elicit higher levels of perceived understanding and no increase in system-like trust in the algorithm compared to no presentation. H2 has to be rejected (for descriptive values, see Table 1).

4.4 Combination of Trust-building Measures

Another MANOVA was computed to examine which combination of trust-building measures elicits the highest level of a) perceived, b) factual understanding, and c) system-like trust (RQ1). As dependent variables, the calculation considered perceived understanding, factual understanding, system-like trust and trust-building measures as between-subject factor.

The MANOVA showed a significant difference between different combinations of trust-building measures on the combined dependent variables, $V = 0.11$, $F(9, 1296) = 5.27$, $p < .001$. Post-hoc univariate ANOVAs were calculated for every dependent variable (see Table 2 for summary). Results showed a significant effect on perceived understanding, supporting RQ1a, and a significant effect for factual understanding, supporting RQ1b. However, there was no effect on system-like trust, which opposes RQ1c.

Post-hoc tests (Tukey-HSD) were considered to identify the trust-building measure that achieved the highest levels of perceived understanding and factual understanding. Tukey HSD post-hoc analysis on perceived understanding revealed a significant difference between the XAI group and the AI-Seal group, $p = .022$ (*M_Diff* = 0.25, 95%-CI[0.02, 0.47]), but not between the XAI group and combination group, $p = .968$ (*M_Diff* = 0.04, 95%-CI[-0.18, 0.26]), nor between the AI-Seal group and the combination group, $p = .073$ (*M_Diff* = -0.21, 95%-CI[-0.43, 0.01]). Post-hoc tests on factual understanding showed a significant difference between the XAI group and the AI-Seal group, $p < .001$ (*M_Diff* = 0.73, 95%-CI[0.26, 1.20]), and the AI-Seal group and combination group, $p < .001$ (*M_Diff* = -0.98, 95%-CI[-1.47, -0.51]), but not between the XAI group and the combination group, $p = .511$ (*M_Diff* = -0.25, 95%-CI[-0.71, 0.21]). This suggests that the highest level of perceived understanding was achieved by XAI, whereas the highest level of factual understanding was achieved by XAI and the combination (Table 1).

5 Discussion

As AI becomes more prevalent in daily life, trust-building measures have emerged to improve users' trust and understanding. This study examines the effectiveness of these measures in improving users' factual understanding, perceived understanding, and system-like trust in AI. The results reveal that XAI had a positive impact on factual understanding and perceived understanding (H1), while an AI-Seal had no effect on these constructs (H2). XAI performed best in terms of perceived understanding, whereas XAI and the combination with an AI-Seal performed best in terms of factual understanding (RQ1).

5.1 Effect of XAI

The finding that XAI leads to greater understanding has been repeatedly shown (e.g., [13]), and our study was able to confirm this, suggesting that explanations indeed help individuals develop a correct mental model of how AI works, translating into greater understanding [10, 39].

Considering the relationship of trust and XAI, previous findings led us to hypothesize that XAI should increase system-like trust (e.g., [9, 10]). However, since system-like trust was not significantly increased by XAI, we cannot confirm these results. A possible explanation could be, as there is no homogeneous research which type of XAI leads to the highest trust [42], another type of explanation might have yielded different results. Another reason could be that our trust measurement was just one of many possible [44]. A different, for instance, longer scale with more items might have yielded different results. Moreover, the breakfast guide used in this research may have been overly simplistic, as it only provided recipe recommendations and lacked more complex suggestions, such as those generated by decision support systems in high-risk scenarios like medical diagnoses. Scholars like Schraagen et al. [37] contend that the context in which an XAI is implemented also plays a crucial role in fostering trust in AI. Therefore, an XAI employed in a more complex setting may have generated more trust.

In summary, XAI contributes to a better understanding of AI, but does not necessarily increase trust. The assumption that understanding simultaneously leads to trust [10] cannot be confirmed and raises the question if XAI should be regarded as an "understanding-building" rather than a "trust-building measure".

5.2 Effect of AI-Seal

The results of H2 show that neither perceived understanding nor system-like trust increased due to an AI-Seal. Although the argument that an AI-Seal might be a good measure to generate trust and perceived understanding in the future [36, 41, 43] could not be confirmed, this might be different for an already established AI-Seal.

A possible explanation lies in the broader contextual and cognitive conditions under which trust in certification labels emerges. A critical factor is the absence of an established, widely recognized AI-Seal, which can lead to uncertainty or skepticism in users who lack sufficient prior knowledge to assess its credibility. This aligns with findings by Wischnewski et al. [43], who showed that trust in an AI system increased only when participants also trusted the issuing institution behind the seal, suggesting that institutional credibility and background transparency are critical factors in shaping the effectiveness of certification measures. Furthermore, Scharowski et al. [36] found that certification labels tend to increase trust particularly in high-stakes contexts, indicating that the relatively low-stakes breakfast scenario used in this study may not have been sufficiently relevant to trigger comparable trust effects.

In terms of user understanding, the AI-Seal likewise showed no significant effect on perceived understanding. This may be due to the limited informational value typically provided by certification labels, which often lack explanatory depth. Without any additional explanation about the AI system, the seal alone likely failed to stimulate meaningful cognitive engagement. This interpretation is further supported by Wischnewski et al. [43], who found that although participants expressed a strong desire for verification seals, they simultaneously reported the lowest levels of understanding of these same seals. This suggests a mismatch between the informational expectations associated with a seal and its actual cognitive utility. Similarly, Burkell [5] argues that seals may lead to a perceived sense of understanding without offering substantial informational value.

Thus, the findings indicate that an AI-Seal alone may currently be insufficient to foster trust or understanding, particularly when it lacks contextual relevance, institutional credibility, or complementary explanatory support. Taken together, this null-finding is a crucial contribution, as it challenges the assumption that certification is a universally effective tool and highlights the specific boundary conditions under which such heuristic cues may be ineffective.

5.3 Combination of Trust-building Measures

Regarding RQ1, results revealed that presenting XAI led to the highest level of perceived understanding, while the combination of XAI and an AI-Seal as well as XAI alone, resulted in the highest level of factual understanding. Additionally, no effect of presenting a trust-building measure on system-like trust could be found.

The finding that both factual understanding and perceived understanding are higher when using XAI compared to presenting an AI-Seal suggests that XAI may have encouraged more in-depth cognitive processing of the system's functioning, whereas the AI-Seal did not necessarily act as an effective heuristic cue for trust or understanding. Contrary to findings by Buçinca et al. [4], who argued that explanations may serve as heuristics rather than prompting deeper reasoning, our results point to a more elaborative processing of XAI in this context. While this suggests that XAI could be seen as most effective, in terms of understanding, the AI-Seal did not have a negative impact on the effectiveness of XAI, as seen in the combination condition, and could potentially enhance the effect of XAI on understanding by directing more attention to XAI, since the combination achieved slightly higher mean values regarding factual understanding than XAI alone (Table 1).

Although AI-research suggests that the key to increasing trust in AI is by creating more transparency and understanding about how AI works [40], we could not confirm this as no trust-building measure had an effect on system-like trust. However, since this was already seen and discussed, in H1 and H2 regarding XAI and AI-Seal, it seems plausible that combining the two methods does not significantly affect the users' system-like trust.

The question of which of the trust-building measures presented should now best be used in the future to increase trust in and understanding of AI-applications can therefore still not be answered unequivocally.

5.4 Limitations and Future Work

While the study offers valuable insights into trust and understanding in AI systems, some limitations point to avenues for future research. First, although validated instruments were used, alternative operationalizations might have influenced the results. The field would benefit from further development of standardized and context-sensitive measures, ideally supported by meta-analyses and cross-study comparisons [44].

Second, the chosen application scenario – a breakfast recommendation system – ensured experimental control but may not reflect the complexity and perceived risk of more consequential AI systems. As Scharowski et al. [36] note, certification effects tend to be more pronounced in high-stakes contexts. Future studies should therefore investigate trust-building measures in domains such as healthcare, finance, or mobility, where users may process interventions like AI-Seals differently.

In addition, the AI-Seal may not have been sufficiently salient, particularly in the initial subsample. Despite improved implementation, limited recognition could have reduced its potential impact. This aligns with prior research suggesting that unfamiliar visual cues may be processed heuristically or subconsciously, especially when not supported by institutional framing [29].

Despite these limitations, AI-Seals remain a promising avenue for establishing AI systems in society if society is informed about who certifies AI and according to which rules. Future research should therefore examine how transparency about certifying bodies and underlying criteria influences their effectiveness. However, it also remains to be investigated whether a seal alone can lead to more trust and understanding in a system as complex as AI, or whether it is rather perceived as an expression of quality.

Additionally, the combination of XAI and AI-Seal yielded the highest mean scores in terms of factual understanding, indicating that further investigation of this approach may be a promising contribution towards the black-box problem. Building on this, future work could explore additional types of XAI, such as interactive explanations [1], as well as alternative strategies to foster trust and understanding - such as improving AI literacy in the broader population.

5.5 Conclusion

This study contributes to the ongoing discourse on fostering trust and understanding in AI systems, emphasizing the need for further research on building understanding and trust in AI. While previous research has suggested that XAI holds potential for creating understanding of and trust in AI systems [9, 39], our results only show an increase of factual and perceived understanding, challenging the common assumption that understanding leads to trust [10]. The AI-Seal, by contrast, showed no significant effect on either system-like trust or perceived understanding, highlighting critical boundary conditions – such as institutional credibility and contextual relevance – that challenge the standalone efficacy of such certification mechanisms. As for the combination of the XAI and the AI-Seal, the results contradicted the assumption that it may be a trade-off for the disadvantages of each method in order to maximize the advantages, by only showing a significantly higher factual understanding.

Taken together, the findings underscore the importance of critically examining both explanatory and symbolic strategies in AI communication. There is a need for more nuanced research into how different trust-building measures interact, and how character-based factors influence user evaluations. Ultimately, the goal remains to design AI systems that are not only functionally effective but also transparent, trustworthy, and accepted within society.

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